Oliver Nemec

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Experience

Datadog New York, NY

Software Engineer II *Dec 2025 – Present*

Software Engineer *July 2024 – Dec 2025*

- Architected an enterprise Just-In-Time (JIT) Privileged Access Platform on Temporal to eliminate standing super-admin access, cutting data models 40% across 87 RFC reviews; built a durable state machine (TTL timers, signal revocation) fronted by OpenAPI Go/CLI clients, unifying scattered Lambdas into an extensible platform that consolidates onboarding, offboarding, license management, and inactivity auditing.
- Led company-wide deprecation of legacy directory services handling millions of monthly calls (>100K daily) across >120 call sites with zero downtime; mapped monorepo call chains via Static Single Assignment (SSA) call graphs, sequenced cross-team migration roadmaps, and deployed Datadog dashboards tracking GraphQL, Temporal, and SQL access patterns.
- Scaled an enterprise AI Slack assistant from ~5 to >100 teams while driving zero-downtime AI Gateway platform migration; re-engineered the user-feedback data pipeline (Postgres to S3/Snowflake) using incremental diff-based synchronization, slashing runtime 99% (15+ min to <5s) and memory usage 96% (~7.5GB to ~300MB).
- Eliminated recurring 60-second ALB timeouts across 4 production services by architecting a Kubernetes microservice that containerized the 1Password CLI (`op`) into an authenticated API with OpenAPI-generated Go/Python clients; cut operation runtimes from ~10 minutes to <1 second and delegated downstream service rollouts to a mentored engineer.
- Eliminated ~15,000 monthly CI failures across 57+ repositories to zero by isolating broken commit-author metadata in an unowned shared GitLab CI template; resolved internal package deadlocks by compiling deterministic `uv` lockfiles from `pyproject.toml`, and served as Incident Commander resolving 7+ internal operational and SSO outages.

Software Engineer Intern *May 2023 – Aug 2023*

- Resolved technical debt, improving developer QoL for 3+ teams and achieving 99% AWS cost savings through workflow optimizations.
- Implemented static analysis in GitLab CI and Pydantic models for HTTP/API requests, reducing errors by 20% and data-validation latency by 60%.

PathAI Boston, MA

Software Engineer Co-op *May 2022 – Dec 2022*

- Built an ETL pipeline using Kubernetes and Apache Airflow, migrating data from 5+ sources into Snowflake, enabling unified data access and analytics for cross-functional teams.
- Designed a custom Helm chart for Airflow on EKS and integrated REST APIs, ensuring scalable and reliable data orchestration.

Education

Northeastern University Boston, MA

B.S. Computer Science (ML/AI Concentration) *Jan 2021 - May 2024*

Technical Skills

Languages: Python, Go, SQL (PostgreSQL), Java, C/C++
Cloud & Infrastructure: AWS (Lambda, S3, EventBridge), Kubernetes, Docker, Helm, Terraform, GitLab CI
Systems & Frameworks: Temporal (Workflow Orchestration), REST / OpenAPI, gRPC, GraphQL, Apache Airflow, Snowflake
Tooling & Architecture: Static Single Assignment (SSA) Analysis, `uv`, Datadog (APM, Tracing, Metrics), Git

Projects

Sleepypod | *Embedded Linux, Python, CBOR, NATS, Vitest, systemd* | GitHub 2024 – Present

- Contributed to open-source embedded hardware platform: resolved CBOR telemetry stream crashes (v2.5.0), modeled TEC overcurrent pulses (>11.5A) and fan PWM thresholds in Vitest, and documented surgical firmware flashing across NXP/MediaTek SoCs and eMMC A/B slot failover.

Thermal Printer | *Python, Playwright, ESC/POS, `uv`* | GitHub Sept 2023 – Present

- Architected an asynchronous IoT printing daemon converting dynamic HTML templates into rasterized 1-bit bitmap graphics via headless Playwright, communicating directly with thermal hardware over ESC/POS USB protocols with `uv` package isolation.